

SYMMETRICALLY FATED PROGENITORS DYNAMICALLY ACCOMMODATE TISSUE MAINTENANCE IN THE ESOPHAGUS (GEO Accession: GSE255401)

INTRODUCTION

The purpose of this study is to clarify the mechanism underlying the maintenance of the esophageal epithelium tissue, in particular it focuses on a particular subpopulation of progenitor cells marked by **Troy**, a **cell surface receptor**, which display distinct characteristics compared to other progenitors Troy negative.

While many epithelial tissues are maintained by long-lived stem cells, in vivo experiments using lineage tracing and mathematical models, have suggested that the esophageal epithelium is sustained by the proliferation of relatively short-lived progenitor cells located in the basal layer. As they differentiate, these cells move away from the basal layer and ascend into the upper layers losing their ability to divide and acquiring the morphological characteristics typical of squamous epithelium.

These progenitor cells preserve the homeostasis of the stratified epithelium by making stochastic decisions about their fate. Each cell division can result in two progenitor cells or two differentiated cells, but symmetric outcomes are very rare, or one progenitor and one differentiated cell.

Troy, in fact, works not only as a **marker of progenitor cells** but also as an active regulator, a **negative regulator of proliferation**. This suggests that Troy may help prevent excessive expansion of progenitor cells, thereby maintaining a proper balance between renewal and differentiation.

Using single-cell sequencing, isolating this subpopulation by FACS sorting, researchers discovered that Troy was **highly present in undifferentiated basal cells clusters**, slightly present in committed to differentiation progenitor clusters and rarely present in differentiated cells.

RNA-SEQUENCING EXPERIMENT

Based on these observations researchers wanted to understand how Troy is able to act like a negative regulator of proliferation, its role in differentiation, its involvement in the WNT pathway and its role in tissue stress conditions.

- **Samples:**

Researchers used **Troy^{CreERT2}; Rosa26^{tdTomato} mice**, this system allows to activate the Cre recombinase under the control of Troy promoter only in the presence of tamoxifen, so that only the cells expressing Troy at the moment of tamoxifen subadministration will become red fluorescent (**Troy tdTomato⁺**). Cells not expressing Troy will be named **Troy tdTomato⁻**.

They tested both Troy WT/KO and Troy KO/KO mice:

	tdTomato+	tdTomato-
Troy WT/KO	Cells expressing Troy	Cells that don't produce Troy
Troy KO/KO	Cells not expressing Troy	Troy-null cells

Total number of samples: 16

- **Technique:**

Bulk RNA-seq on progenitor cells isolated by FACS sorting.

- **Platform:**

Illumina NovaSeq 6000

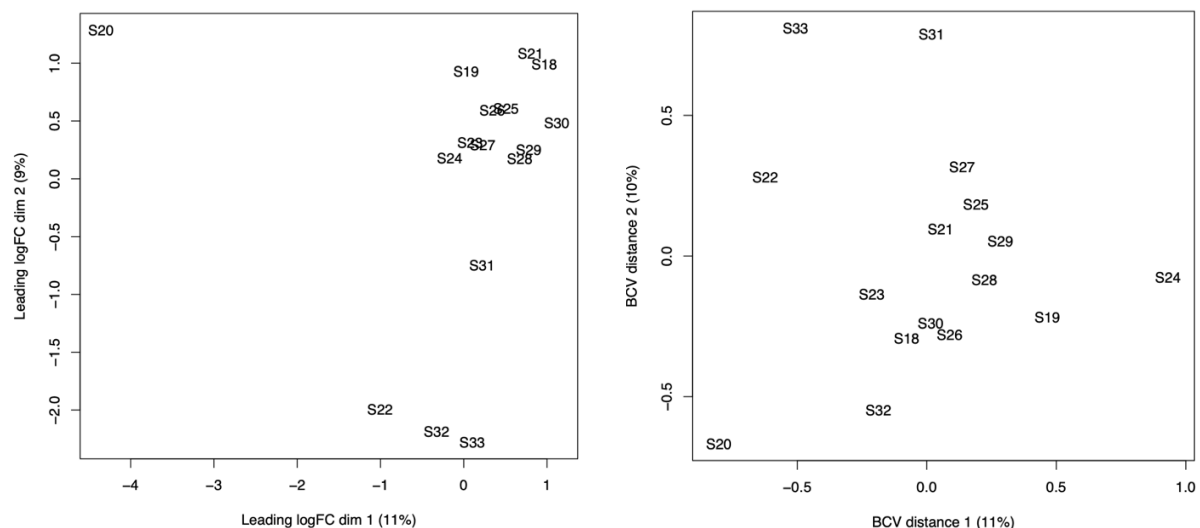
RESULTS FROM R

Differential Gene Expression Analysis (EdgeR)

- MDS Plots

These plots shows the similarity between different data points. The first plot is based on the log Fold Changes, so it plots samples based on the genes with the largest changes in expression. In this case S20, S22, S32 and S33 seem to be separated from others, suggesting that the genotype or the Troy status are affecting gene expression.

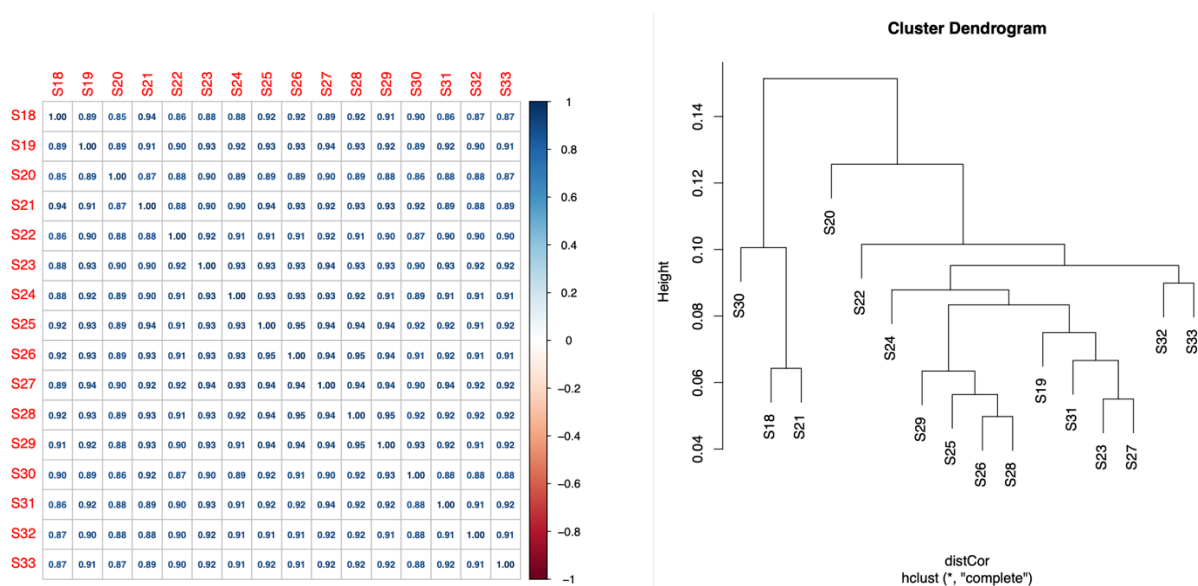
The second plot is based on the BCV distances and it displays how biologically variable the samples are relative to each other. S20 and S22 again lie far from the main cluster. S31 and S33 form a close pair, potentially representing a Troy-negative signal.



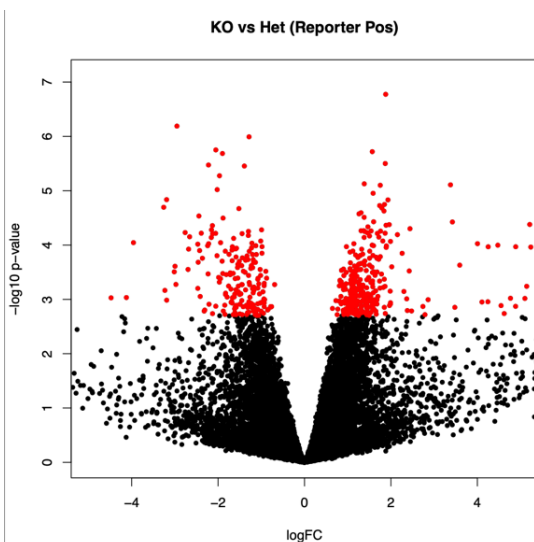
- Correlation matrix and Hierarchical Clustering dendrogram

This Correlation matrix calculates the Pearson correlation coefficient between two samples and composes a grid. Values close to 1 mean a high similarity between the considered samples in their expression profile. The matrix confirms samples correlate quite well with each other except for S20, S22 and S32 which display reduced correlations with several samples, potentially reflecting a different transcriptional profile.

The Cluster Dendrogram groups samples by similarity using a distance matrix. It seems there may be slight subrouping like S18 and S21 or S32 and S33 which are very close. Anyway samples are not extremely separated suggesting any strong outliers is present.

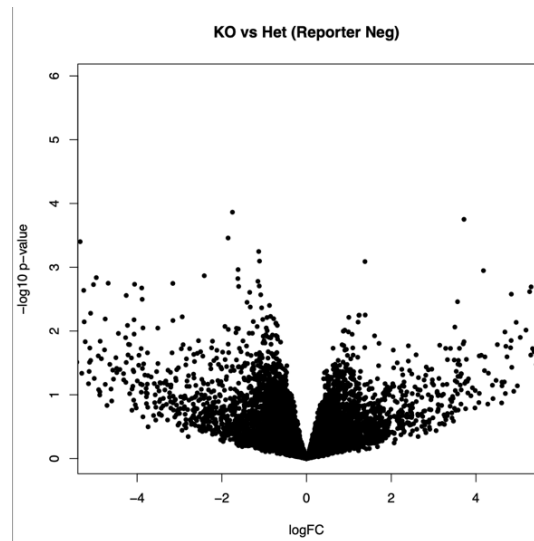


- Volcano Plots

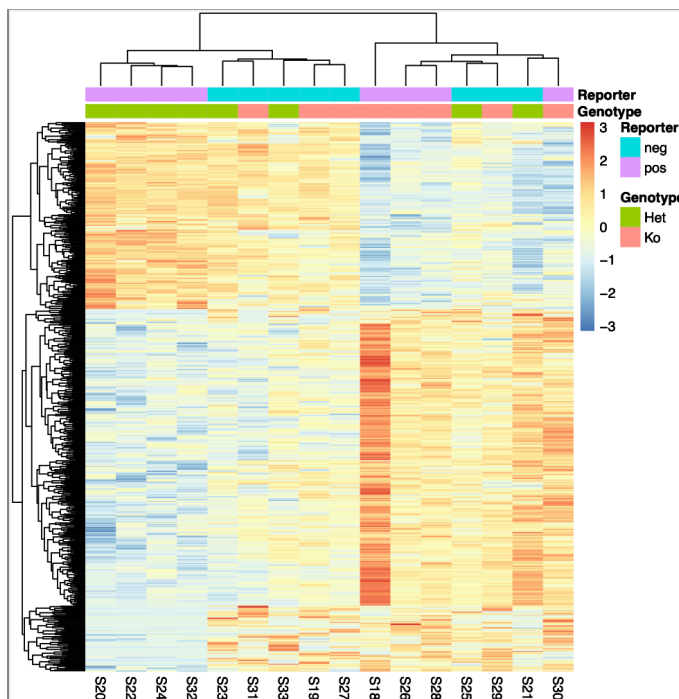


I have found ~ 560 significant differentially expressed genes with $FDR < 0.05$. This means Troy has a clear and strong effect on tdTomato+ progenitor cells.

In the case of deletion of Troy in tdTomato-progenitors any significant differentially expressed gene is identified. This confirms that the Troy effect is cell-specific: it acts only where it is expressed and Troy KO doesn't impact cells which are already Troy negative.



- Heatmap



The Heatmap confirms the above findings, showing that Troy knockout affects only tdTomato-positive cells, while tdTomato-negative cells remain unaffected.

Functional Enrichment Analysis

Considering the results of the Differential gene expression analysis I performed the Functional enrichment analysis only on the KO vs Heterozygous tdTomato positive cells contrast, since tdTomato negative one didn't produce any significant result. The EdgeR analysis reported 372 up-regulated genes and 193 down-regulated ones.

For both ORA and GSEA I considered the following categories:

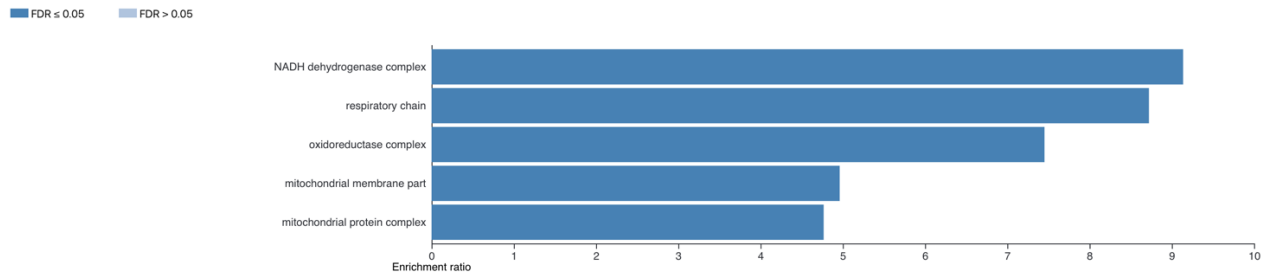
- geneontology_Biological_Process_noRedundant
- geneontology_Cellular_Component_noRedundant

- geneontology_Molecular_Function_noRedundant
- pathway_Reactome

ORA

- **Cellular Component**

DOWN-regulated genes:



NADH dehydrogenase complex: ND1, ND2, ND4, ND5

Respiratory chain: ND1, ND2, COX1, ND4, ND5, ND6, CYTB

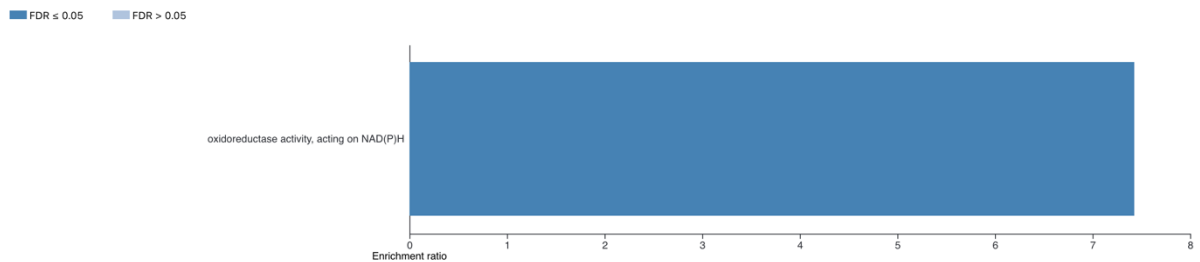
Oxidoreductase complex: Dld, ND1, ND2, COX1, ND4, ND5, CYTB

Mitochondrial membrane part: Ckmt1, Tomm40l, Coq3, Dusp18, ND1, ND2, COX1, ND4, ND5, CYTB

Mitochondrial protein complex: Tomm40l, Mrps25, Mterf4, Mrpl47, 2810006K23Rik, mrpl18, ND1, ND2, COX1, ND4, ND5

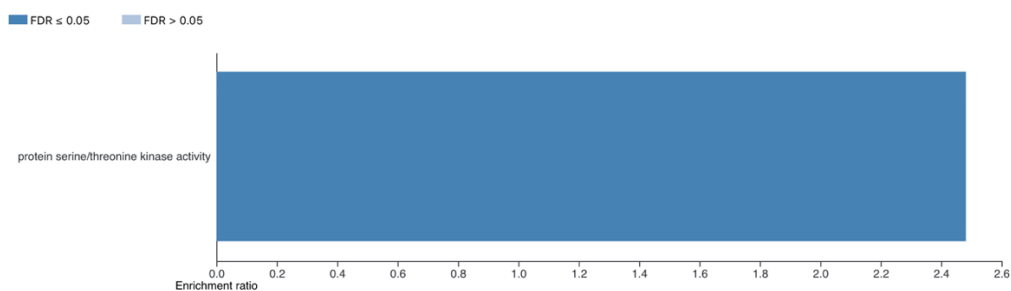
- **Molecular Function**

DOWN-regulated genes:



Oxidoreductase activity, acting on NAD(P)H: Dld, ND1, ND2, ND4, ND5, ND6

UP-regulated genes:



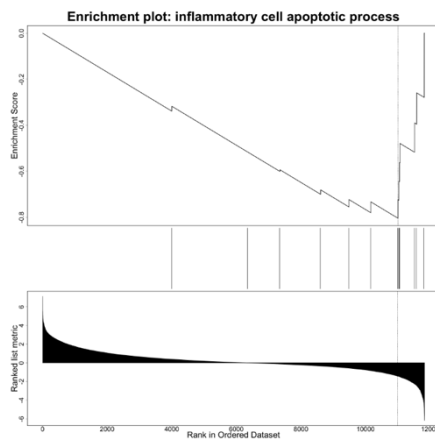
Protein serine/threonine kinase activity: Dyrk1b, Mapk13, Prkacb, Ripk4, Nlk, Mapk12, Pkmyt1, Pask, Stk 25, Nek2, Ercc2, Pak4, Cdk16, Map2k2, Pim3, Mapk6, Adck2, Pskh1, Taok3, Csnk1g3.

GSEA

- **Biological Process**

DOWN-regulated genes:

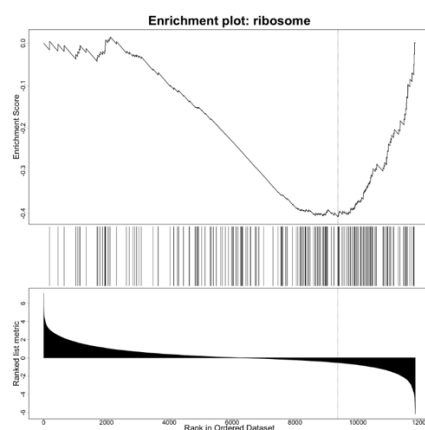
- *Inflammatory cell apoptotic process*; FDR=0.02: Anxa1, Hcar2, Fas, Il18, Sirt1, Zmpste24, Pik3cd



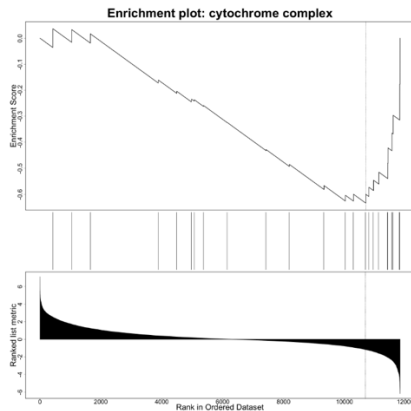
- **Cellular Component**

DOWN-regulated genes:

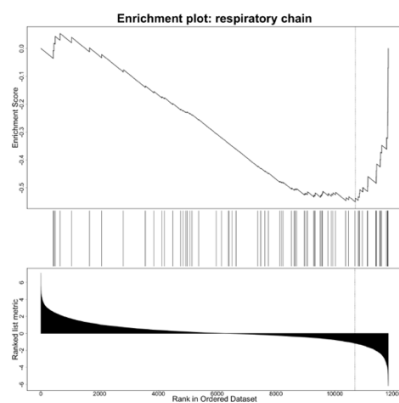
- *Ribosome*; FDR= 0.02: Mrpl47, Mterf4, 2810006K23Rik, Mrpl18, Mrps25, Rpl11, Mrpl55, Zcchc17, Rpl7, Rpl4.



- *Cytochrome complex*; $FDR=0.0049$: COX1, CYTB, Ndufa4l2, Uqcrrh, COX3, Uqcc3, Cox6b1, Uqcrc2, Ndufa4



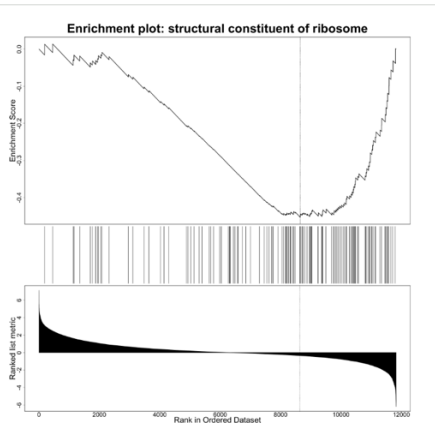
- *Respiratory Chain*; $FDR=2.2e-16$: COX1, CYTB, ND2, ND5, ND4, ND6, Ndufa4l2, Uqcrrh, Ndufa1.



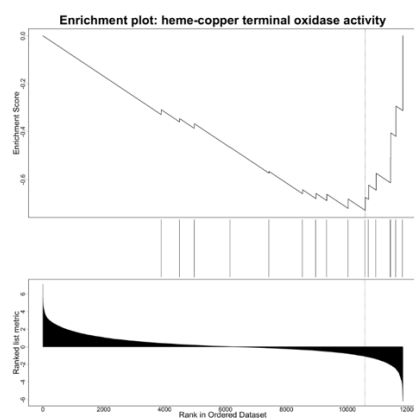
- **Molecular Function**

DOWN-regulated genes

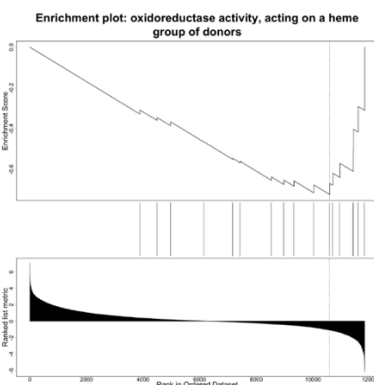
- *Structural Constituent of Ribosome*; $FDR=.0189$: Mrpl47, Mrpl18, Mrps25, Rpl11, Mrpl55, Rpl7, Rpl4, Mrps18b, Rpl9, Mrpl51



- *Heme-Copper Terminal Oxidase Activity; FDR= 0.011*: COX1, Ndufa4l2, COX3, COX2, Cox6c, Ndufa4, Surf1



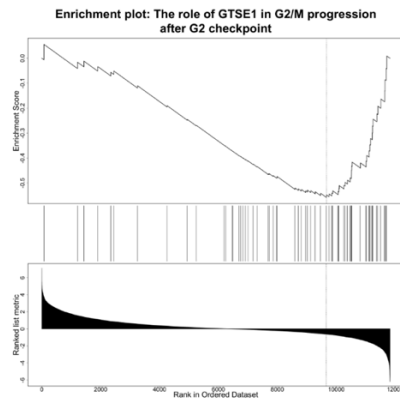
- *Oxidoreductase Activity, Acting on a Heme group of Donors; FDR=0.014*: COX1, Ndufa4l2, COX3, COX2, Cox6c, Ndufa4, Surf1



- **Pathway Reactome**

DOWN-regulated genes:

- *The Role of GTSE1 in G2/M Progression after G2 Checkpoint; FDR=0.023*: Tuba1b, Tuba1c, psmc6, Hsp90aa1, psmc2, trp53, Psmd4, Psma3, tuba4a, Psmd14



CONCLUSIONS

The results obtained from the differential gene expression analysis and the functional enrichment analysis suggest some interesting conclusions that confirm the importance of Troy in the esophageal progenitor cells and its involvement in the maintenance of tissue health and homeostasis.

First of all, we can affirm that Troy **acts as a negative regulator of proliferation** based on the up-regulation of genes involved in proliferation and survival pathways, such as protein kinase genes. In particular the GTSE1 pathway, involved in G2/M transition, is altered in KO cells. It also seems to **act as a regulator of differentiation** since the down-regulation in KO cells of mitochondrial-associated genes which implies a reduced mitochondrial activity leading to a decreased ATP production, a reduced differentiation ability and potential dysfunction or metabolic shift. This suggests that Troy normally keeps proliferation under control avoiding an excessive progenitor expansion, supports mitochondrial activity and respiration maintaining a correct balance between differentiation and cell division.

Moreover, the down-regulation of genes involved in cell apoptotic process such as Anxa1 or Hcar2 underline the ability of Troy to **promote inflammatory apoptosis** so it is able to make cells more susceptible to apoptosis during inflammatory or stress conditions.

Finally, we can declare that Troy is responsible for the **activation of protein synthesis** which appears reduced in KO progenitors due to the down-regulation of ribosomal genes (Mrpl47, Mrpl18, Rpl9, Rpl14...) which supports the hypothesis of a metabolic dysfunction and reprogramming.

BIBLIOGRAPHY

Grommisch D., Wang M., Eenjes E., Svetličič M., Deng Q., Giselsson P., Genander M. (2024): **Defining the contribution of Troy-positive progenitor cells to the mouse esophageal epithelium.** *Dev Cell.* 2024 May 20;59(10):1269-1283.e6.